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	Annual revenue	Net Profit
Acme Computer	138	8
Nadir Software	87	5
Quixote Media	111	13

a) Quixote had largest annual revenue

F

b) Nadir had lowest net profit and Acme Computer had largest annual revenue

T

c) Acme Computer had the largest net Profit or Quixote had the largest net Profit

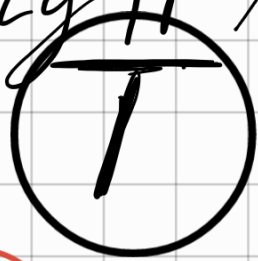
T

d) If Quixote had the smallest net Profit then Acme had the largest annual revenue

T

-?

e) Nadir had the smallest net Profit if and only if Acme had the largest annual rev



② P — It is below freezing

q — It is snowing

a) It is below freezing and snowing.

$P \wedge q$ (P and q)

b) It is below freezing but not snowing

$P \wedge \neg q$

c) It is not below freezing and it is not snowing

$\neg P \wedge \neg q$

d) It is either snowing or below freezing

$q \vee P$



e) If it is below freezing it is also snowing
 $P \rightarrow Q$

f) Either it is below freezing or it is snowing
but it is not snowing if it is below freezing
 $P \oplus Q$ -? $(P \vee Q) \wedge (P \rightarrow \neg Q)$

g) That it is below freezing is necessary
and sufficient for it to be snowing
 $Q \leftrightarrow P$ -? $P \leftrightarrow Q$

③ a) $P \rightarrow \neg P$ 2^n $2^1 = 2$

b) $(P \vee \neg r) \wedge (q \vee \neg s)$ $2^4 = 16$

c) $q \vee p \vee \neg s \vee \neg r \vee \neg t \vee u$ $2^6 = 64$

d) $(P \wedge r \wedge t) \leftrightarrow (q \wedge t)$ $2^4 = 16$

4

a) $p \wedge \neg p$

b) $p \vee \neg p$

p	$\neg p$
T	F
F	T

$p \wedge \neg p$
F
F

$p \vee \neg p$
T
T

c) $(p \vee \neg q) \rightarrow q$

d) $(p \vee q) \rightarrow (p \wedge q)$

p	q	$\neg q$	$(p \vee \neg q) \rightarrow q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	F	T	T
T	F	T	F	F
F	T	F	T	F
F	F	T	F	T?